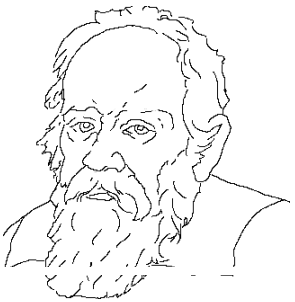
Research has an impact on the way scientists think, what happens in our society and in our environment. Many scientists throughout history contributed including Sir Isaac Newton, Jonas Salk, Orville and Wilbur Wright, Griafielmus Marconi, and Galileo Galilei.

Scientists think, study, wonder why things happen, what caused things to happen, try out ideas, experiment, record their results, share their results, listen to others, and adjust their ideas if they need to change. These are some of the many scientists throughout history that have changed our society, scientific thought, and the environment:

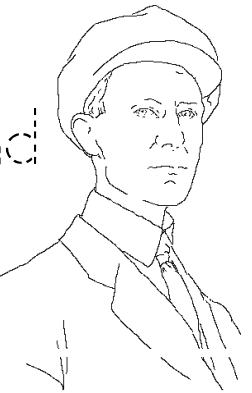
Sir Isaac Newton

Jonas Salk

Galileo



Orville and Wilbur Wright



Contributions of Scientists and their Research

Read the information and trace the dotted words.

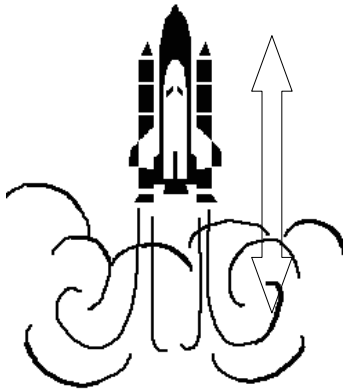
Research means to

observe/watch, record, and test.

Sir Isaac Newton researched gravity and developed the

Three Laws of Motion.

Sir Isaac Newton realized that for every action, there is an equal and opposite reaction. This is Newton's Third Law of Motion.



Newton also studied gravity, which is a force, a pull. Newton determined that objects with greater mass, like the Earth, pull objects with less mass, like an apple, toward them.

Contributions of Sir Isaac Newton
1642-1727

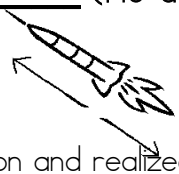
Read the information and trace the dotted words.

If you are doing research you, _____.
(sbereov)

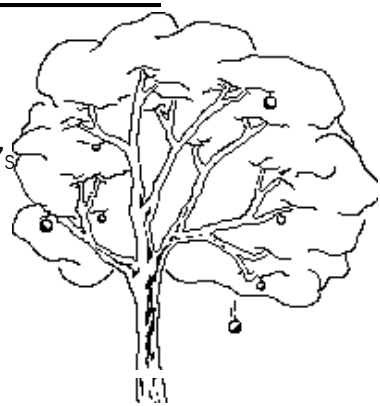
Scientists need to _____ (edrrco) their data (information on what they research).

This scientists studied the effects of gravity and motion.

_____ (riS aslac Notwne)



Sir Isaac Newton studied motion and realized that for every action there is an equal and opposite _____.
(taerncoi)



_____ (vitygra) is the Earth's pull on an object.

Contributions of Sir Isaac Newton

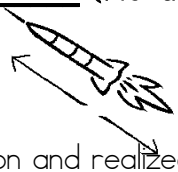
Unscramble the letters to fill in the blanks.

If you are doing research you, _____.
(sbereov)
observe

Scientists need to _____ (edrrco) their data (information on what they research).
record

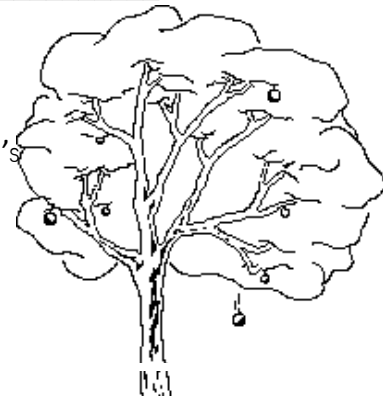
This scientists studied the effects of gravity and motion.

_____ (riS aslac Notwne)
Sir Isaac Newton



Sir Isaac Newton studied motion and realized that for every action there

is an equal and opposite _____.
(taerncoi)
reaction

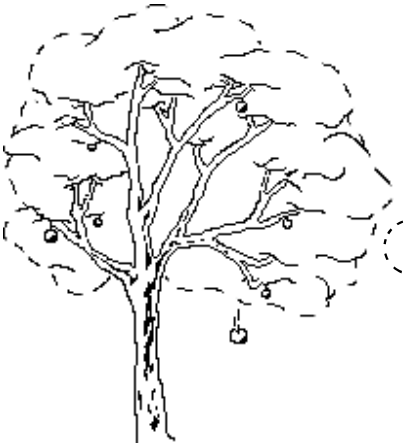


_____ (vitygra) is the Earth's
Gravity

pull on an object.

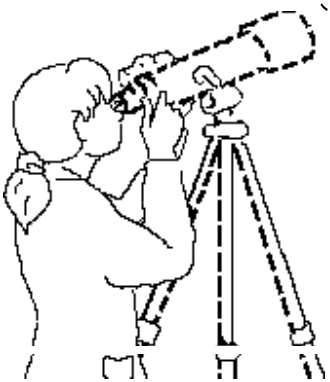
Contributions of Sir Isaac Newton *Answers*

Newton's Third Law of Motion



Gravity

Sir Isaac Newton



Research

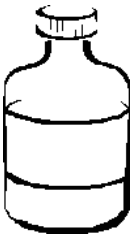
Contributions of Sir Isaac Newton

Trace the words and complete the pictures.

Jonas Salk was an American scientist that did research (observed, experimented, recorded data, and studied results) on polio viruses.

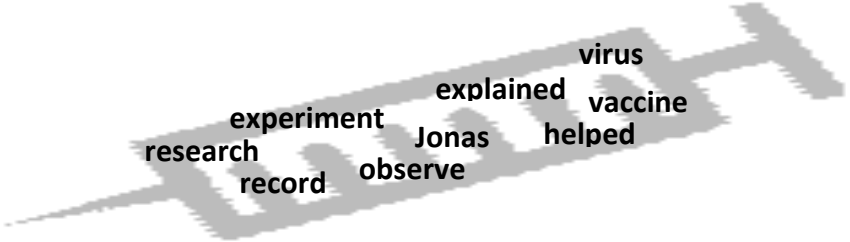
A virus is a disease-causing agent. A virologist studies viruses and the diseases they cause so the scientist can isolate the viruses, grow them, and find ways to stop them from causing the diseases thus finding a vaccine.

Polio was a world-wide disease that was crippling and killing hundreds of people in society and no one knew how it was spread or how to prevent it. Salk's research on the polio virus saved people and prevented the spread of polio in society.



Contributions of Jonas Salk
1914-1995

Read the information and trace the dotted words.



1. When Salk was researching, he needed to O _____,

   study, and read to be able to develop a V _____.

2. Salk needed to O _____ to isolate the polio virus

   and made a R _____ of his observations.

3. The polio V _____ made over 300,000 people in society sick with the disease called polio.

4. Salk explained _____ how the polio virus was transmitted (spread) in society.

5. J _____ Salk was a virologist whose

   res _____

   h _____ society.

Contributions of Jonas Salk

Use the words in the word bank to fill in the blanks.

Any scientist that studies viruses	research
A polio virus causes a disease called	vaccine
The scientist who researched and developed a vaccine for polio was	Jonas Salk
A substance that keeps people in a society from getting a disease, like polio, is a	virologist
When you read, study, experiment, and record your data, you are doing	polio

Contributions of Jonas Salk

Match the word phrase with the correct dotted word.

Any scientist that studies viruses

A polio virus causes a disease called

The scientist who researched and developed a vaccine for polio was

A substance that keeps people in a society from getting a disease, like polio, is a

When you read, study, experiment, and record your data, you are doing

research

vaccine

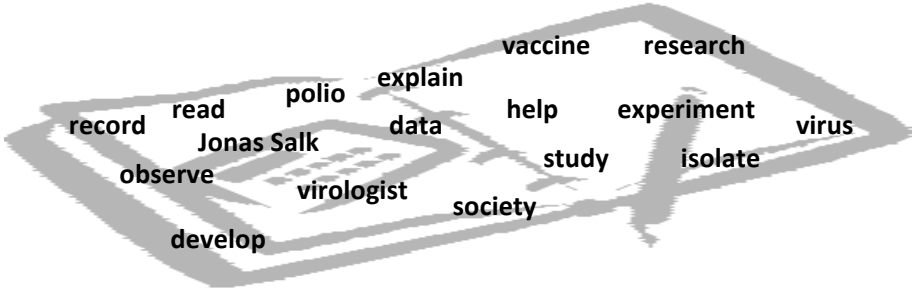
Jonas Salk

virologist

polio

Contributions of Jonas Salk

Answers



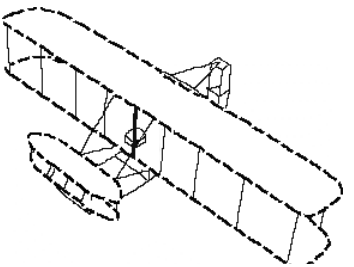
V	A	C	C	I	N	E	Q	W	E	R	I
I	H	J	K	L	P	O	I	U	Y	T	S
R	E	S	E	A	R	C	H	Z	X	V	O
O	V	B	N	M	L	K	J	H	G	F	L
L	D	S	D	A	Q	E	T	U	O	L	A
O	B	S	E	R	V	E	J	G	D	A	T
G	X	V	V	E	N	X	K	E	H	F	E
I	S	W	E	C	R	P	Y	X	J	I	P
S	S	I	L	O	H	E	L	P	O	C	S
T	E	M	O	R	P	R	E	L	N	R	T
Y	R	A	P	D	N	I	U	A	A	S	T
E	M	P	U	S	S	M	F	I	S	G	I
T	A	M	I	T	N	E	O	N	S	D	O
M	V	P	I	U	N	N	R	E	A	D	I
M	I	O	I	D	A	T	A	A	L	M	O
R	R	L	E	Y	M	O	R	D	K	E	M
E	U	I	I	N	N	O	M	I	N	I	P
A	S	O	C	I	E	T	Y	T	R	I	E
T	F	I	L	I	L	I	S	A	N	C	T
Q	W	E	R	T	Y	U	I	O	P	L	K

Contributions of Jonas Salk

Find and circle all the words from the word bank.

Orville and Wilbur Wright were brothers, scientists, and owners of a bicycle shop who wanted to be able to fly. They researched all the available material on flight and made observations about birds. They made several gliders and kites to experiment and collect data.

Using this information they created a flying machine, called the "Flyer". The Flyer was powered by a small gas engine, which they made in their bicycle shop. On December 17, 1903, at Kitty Hawk, North Carolina, the Wright brothers were the first to compete a manned, engine-powered, heavier-than-air flight. In the fourth and longest of the day's four attempts, they flew for 59 seconds and 852 feet (260 meters).



Contributions of Orville, 1871-1948, and Wilbur, 1867-1912, Wright

Read the information and trace the dotted words and picture.

The Wright brothers continued their research and made many improvements to their aircraft. They made several versions of the Flyer and eventually taught others how to build and fly their airplanes. The Wright brothers and their aircraft changed the way society views speed and travel by allowing people to travel great distances in a short amount of time. This developed new industries and provided jobs. Airplanes also affect our environment in bad ways by creating noise and air pollution.

Contributions of Orville and Wilbur Wright

Read the information and trace the dotted words.

1. Orville and Wilbur Wright owned a shop that made •

2. Orville and Wilbur wanted to fly like a •

3. The Wright brothers flew for __ _____ and traveled 852 feet (260 meters). •

4. The Wright brothers' aircraft was called •

5. One of the ways that airplanes positively affect our society is that they allow •

_____

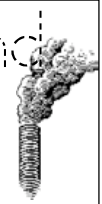
6. Airplanes affect our environment in a bad way by creating •

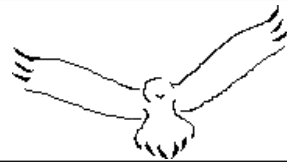
• 59 seconds

• fast travel

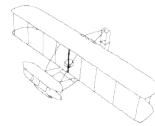
•

• noise and pollution



• 

• *The Flyer*



Contributions of Orville and Wilbur Wright

Match the word phrases with the correct picture or dotted words.

1. Orville and Wilbur Wright owned a shop that made

2. Orville and Wilbur wanted to fly like a

3. The Wright brothers flew for __ _____ and traveled 852 feet (260 meters).

4. The Wright brothers' aircraft was called

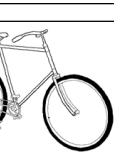
5 One of the ways that airplanes positively affect our society is that they allow

_____

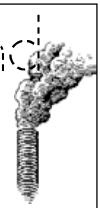
6. Airplanes affect our environment in a bad way by creating

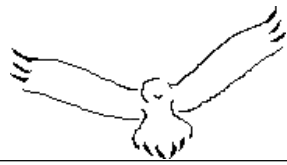
59 seconds

fast travel

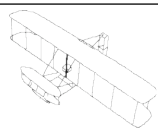


noise and pollution



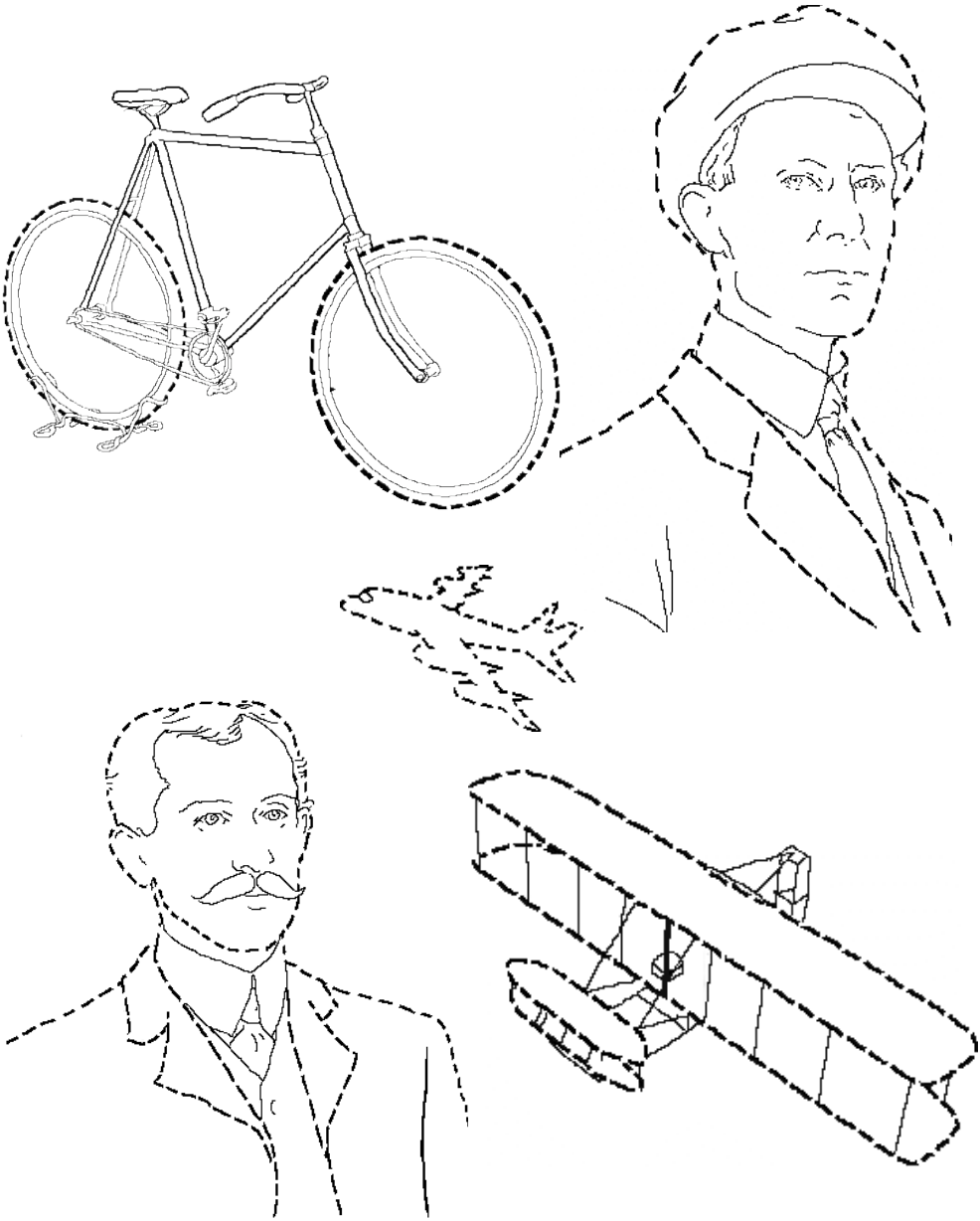


The Flyer



Contributions of Orville and Wilbur Wright

Answers



Contributions of Orville and Wilbur Wright

Trace the dotted lines to complete the pictures.

Marconi was a scientist who did research to develop the wireless telegraph or radio. Marconi sent the first telegraph signal through the air, which developed into all of our wireless communications today: email, television, satellite signals, microwaves Wi-Fi, and cell phones. Until Marconi sent the first telegraph signal through the air, all telegraph messages were sent by wires strung on poles. Marconi sent the first wireless transatlantic (across the Atlantic Ocean) telegraph message.

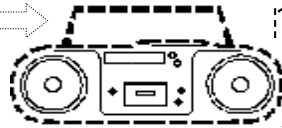
Marconi's research, designs, and development lead to radio broadcasting. He studied short waves and microwaves. Marconi worked on extending signals that could be used day or night. The first transmission from England to Canada in Morse Code was due to Marconi's research.

Contributions of Griafielmus Marconi 1884-1937

Read the information and trace the dotted words.

Marconi's invention of a wireless telegraph developed into the

radio.

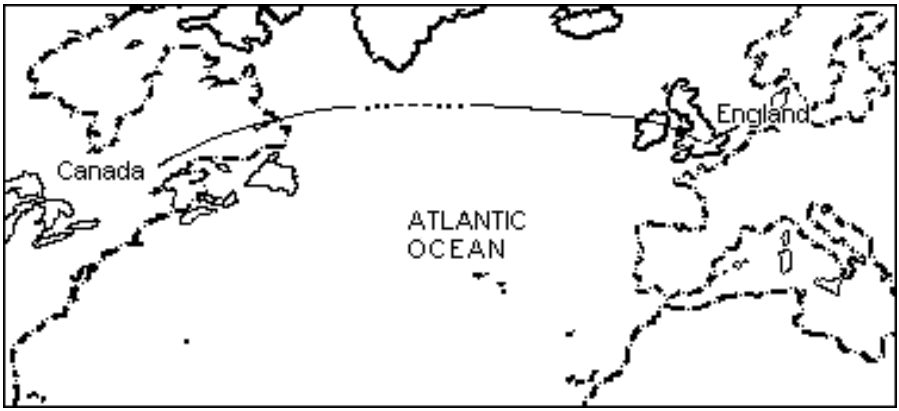


Marconi sent signals through the

air.

Something that crosses the Atlantic Ocean, such as Marconi's wireless telegraph message, is

transatlantic



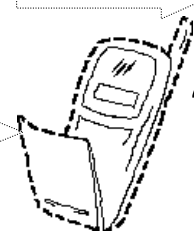
The first transmission was from England to Canada and was sent in

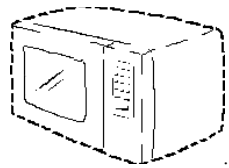
Morse

• • • – – – • • •

Our society uses wireless communication in

cell



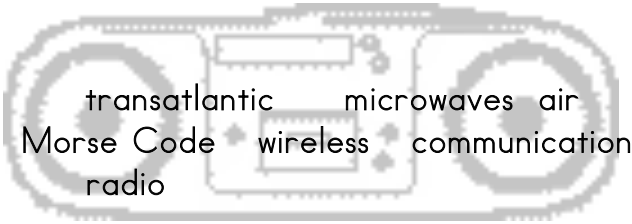


Marconi studied and worked with short waves called

microwaves

Contributions of Marconi

Read the information and trace the dotted words and pictures.



transatlantic microwaves air
Morse Code wireless communications
radio

1. Marconi designed equipment that sent signals through the _____.

2. Another form of wireless communication is a _____.

3. Marconi worked with _____.

4. Marconi made the first _____

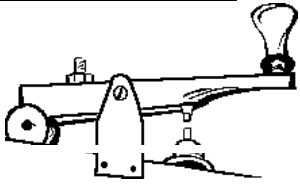
communication from England to Canada.

5. Marconi used _____ _____

with his wireless communication devices and his first transatlantic communication.

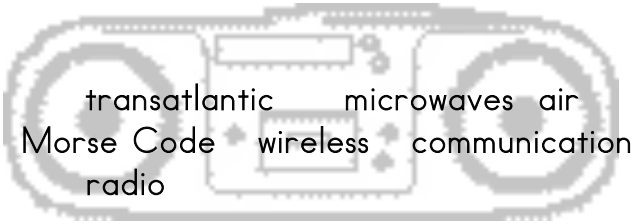
6. Marconi did research on _____

_____.



Contributions of Marconi

Fill in the blanks using the words in the word bank.



transatlantic microwaves air
Morse Code wireless communications
radio

1. Marconi designed equipment that sent signals through the a _____.
air

2. Another form of wireless communication is a r _____.
radio

3. Marconi worked with m _____.
microwaves

4. Marconi made the first t _____
transatlantic

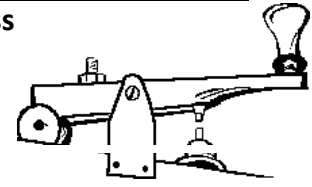
communication from England to Canada.

5. Marconi used M _____ C _____
Morse **Code**

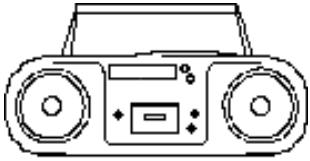
with his wireless communication devices and his first transatlantic communication.

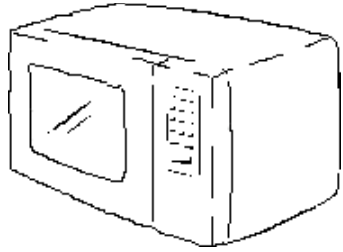
6. Marconi did research on W _____
wireless

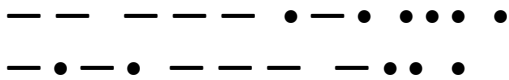
c _____.
communications

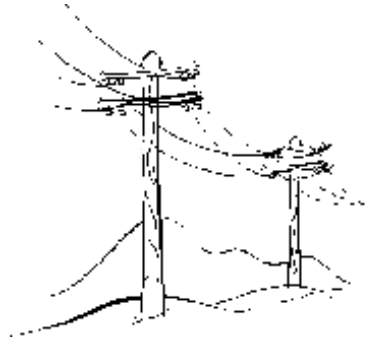


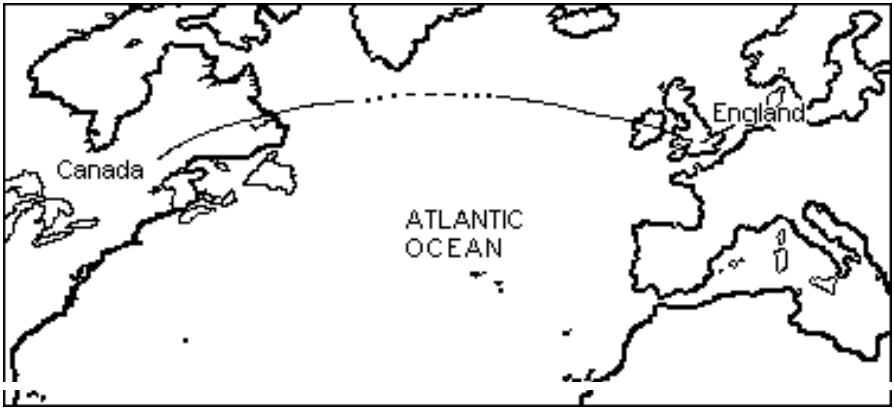
Contributions of Marconi *Answers.*









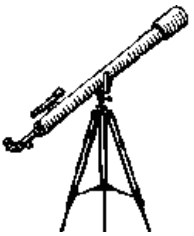


Contributions of Marconi

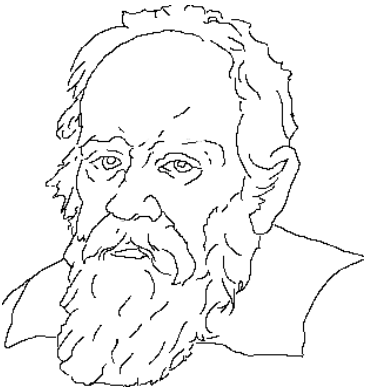
Circle all the things with which Marconi was involved.

Galileo was a scientist in the 17th century who

studied physics, math, philosophy and astronomy. Among Galileo's many contributions to science were his improvements to the telescope. Galileo's telescope allowed him to observe the stars and the planets in our solar system. With it he tracked the phases of Venus. He also discovered Jupiter's four largest moons, which later astronomers called the Galilean moons in his honor.



Galileo did not share the belief that the universe revolves around the earth, but, like a few scientists before him, theorized that the earth and other planets orbit the sun. This made his discovery of Jupiter's moons significant, because they orbit Jupiter, not Earth. The discovery of the phases of Venus proved that Venus orbits the sun.



Contributions of Galileo Galilei
1564-1642

Read the information and trace the dotted words.

orbit revolve philosophy math Jupiter's physics telescope Venus Galileo planets astronomer sun

1. G_____ was an a_____ who studied the stars.

2. Galileo discovered four of J_____ moons.

3. Galileo designed a t_____ that allowed him to make many significant discoveries.

4. Galileo insisted that the S_____ was at the center of the solar system and that the p_____ revolved around it.

5. Re_____ means to go around another object.

6. Galileo studied p_____, m_____, p_____ and astronomy.

7. Galileo used his telescope to track the phases of V_____.

Contributions of Galileo

Read the information, fill in the blanks, and trace the dotted words.

_____ 1. Galileo discovered deep sea animals.

_____ 2. Galileo designed a telescope.

_____ 3. Galileo believed that the earth orbits the sun.

_____ 4. Venus orbits Jupiter.

_____ 5. Revolve means to go around another object.

_____ 6. Galileo was born in the 20th century.

_____ 7. An orbit is an animal that eats grass.

_____ 8. Jupiter's four largest moons are called the Galilean moons.

_____ 9. Telescopes allow astronomers to observe the stars.

_____ 10. An astronomer studies frogs.

Contributions of Galileo

Write T for true or F for false in the blanks.

_____ 1. Galileo discovered deep sea animals.
F

_____ 2. Galileo designed a telescope.
T

_____ 3. Galileo believed that the earth orbits the sun.
T

_____ 4. Venus orbits Jupiter.
F

_____ 5. Revolve means to go around another object.
T

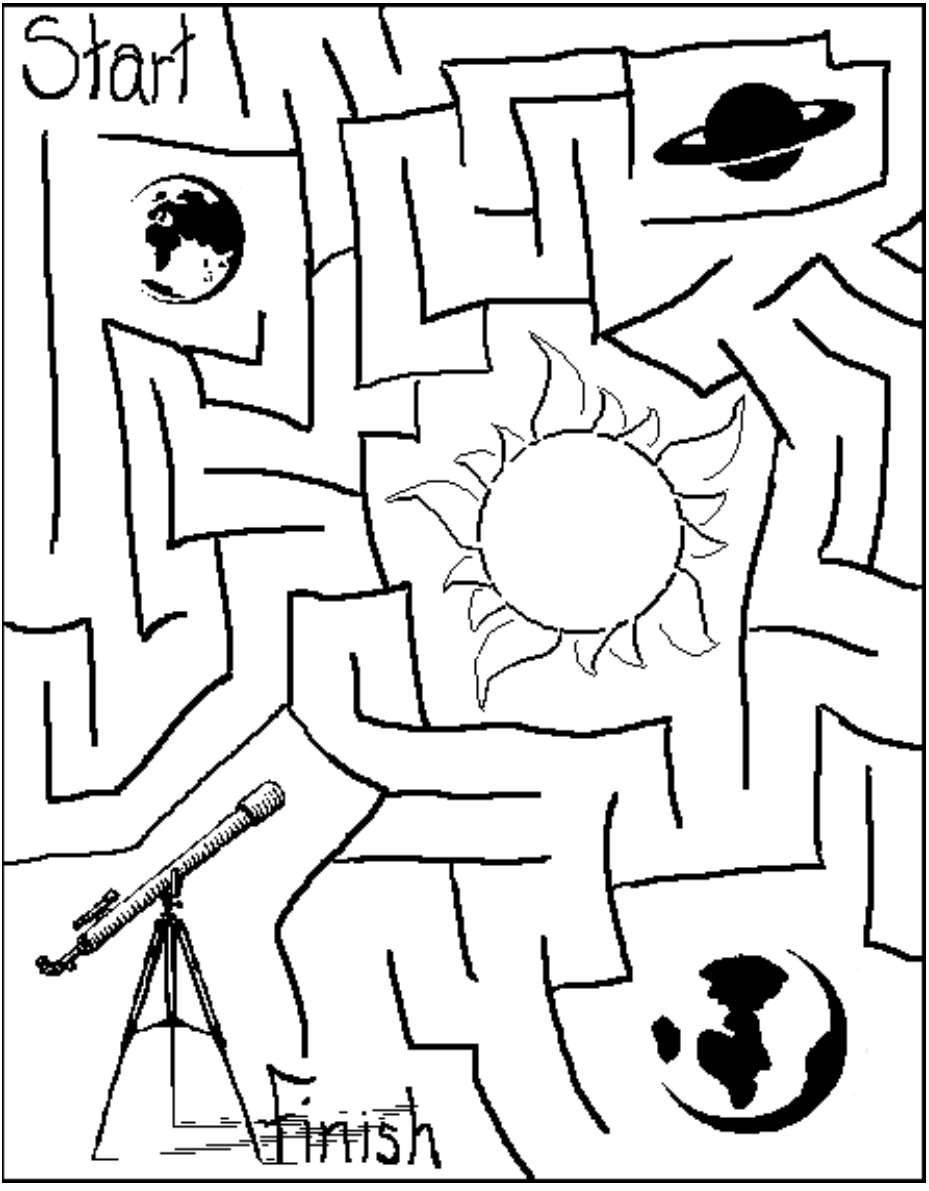
_____ 6. Galileo was born in the 20th century.
F

_____ 7. An orbit is an animal that eats grass.
F

_____ 8. Jupiter's four largest moons are called the Galilean moons.
T

_____ 9. Telescopes allow astronomers to observe the stars.
T

_____ 10. An astronomer studies frogs.
F



Contributions of Galileo

Trace a path to the telescope.

Thomas Edison was a deaf, self-taught scientist whose research and inventions still affect our environment (surroundings) and society today.

His phonograph, which is a device for recording and playing back sound, led to the recording industry giving us records, and eventually cassette tapes, and CD's.

Edison was interested in electricity and used it in much of his work. He was also very persistent, often trying and discarding many ideas before finding one that worked for one of his inventions. He tried many different things before finding a carbon filament for his electric light bulb.

Thomas Edison also invented the motion picture camera and projector, the electric generator, and the electric train.

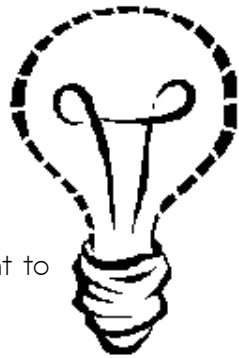
Contributions of Thomas Edison
1847-1931

Read the information and trace the dotted words.

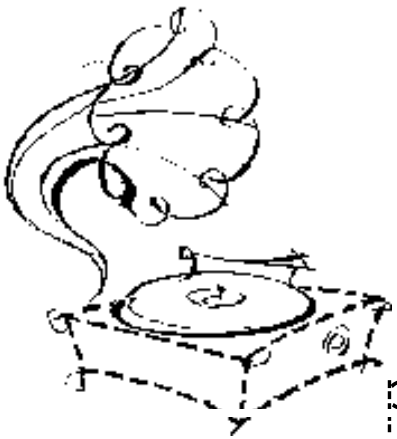
A talented, deaf, self-taught persistent scientist who invented many of the early forms of equipment we use today.

Thomas Edison

Edison tried repeatedly until he found the correct carbon filament to use to make a light bulb light up.



light bulb

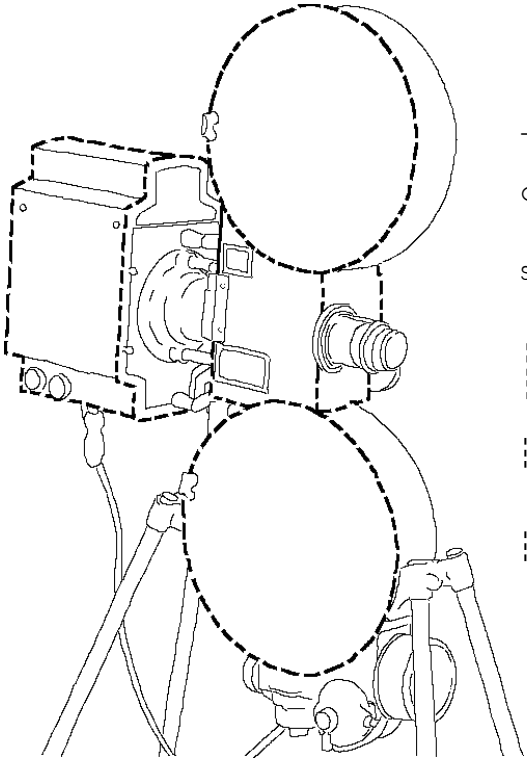


A device invented by Thomas Edison used to record and play back sound.

phonograph

Contributions of Thomas Edison

Read the information and trace the dotted words and pictures.



Thomas Edison invented and designed this equipment to show a series of pictures or make a movie.

environment

The things around you, your surroundings.

persistent

Edison continued to try, over and over again, and never gave up.

Contributions of Thomas Edison

Read the information and trace the dotted words and pictures.

1. Thomas _____ **(nosEdi)** was a deaf, self-taught scientist

   whose inventions changed our society.

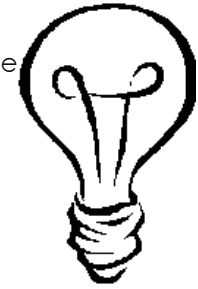
2. Edison invented a machine for recording and playing back sound called

   a _____. **(hngahproop)**

3. Edison experimented with rapid pictures and developed the

   _____ **(niotmo)** picture projector for

   _____ **(viesmo).**



4. Edison was interested in _____. **(lcrctyiitee)**

5. The _____ **(tlghi)** bulb took many attempts to find the

   right element for the filament to give off light.

6. Edison was _____ **(ttenpressi)** in getting his inventions

   to work and would work for days at a time.

Contributions of Thomas Edison

Fill in the blanks with the scrambled words in bold.

1. Thomas _____ (nosEdi) was a deaf, self-taught scientist
Edison

whose inventions changed our society.

2. Edison invented a machine for recording and playing back sound called

a _____. (hngahproop)
phonograph

3. Edison experimented with rapid pictures and developed the

_____ (niotmo) picture projector for
motion

_____ (viesmo).
movies

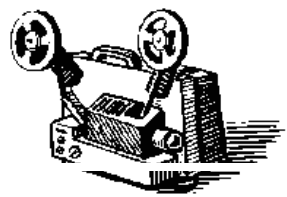
4. Edison was interested in _____. (lcrctyiitee)
electricity

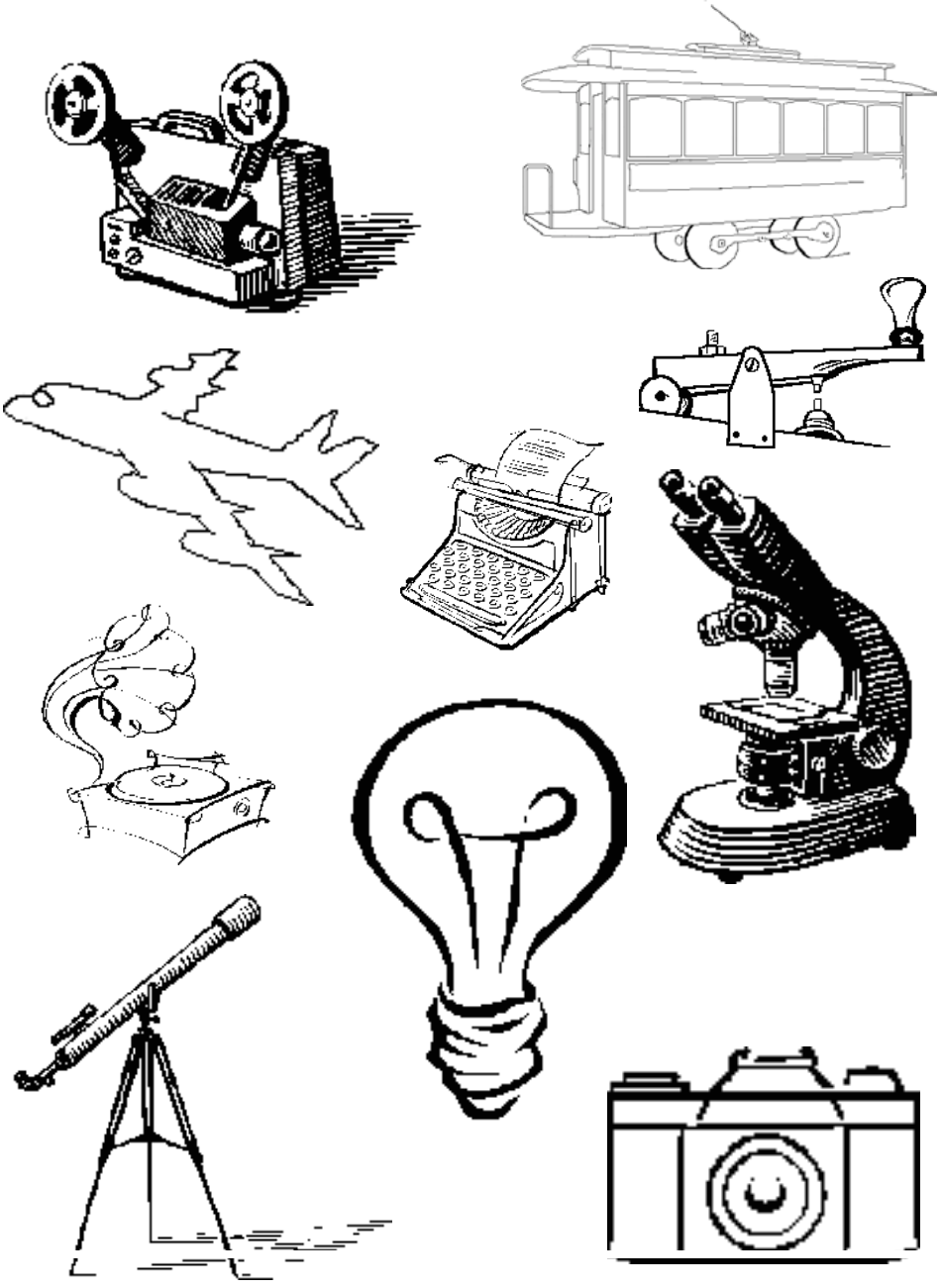
5. The _____ (tlghi) bulb took many attempts to find the
light

right element for the filament to give off light.

6. Edison was _____ (ttenpressi) in getting his inventions
persistent

to work and would work for days at a time.





Contributions of Thomas Edison

Circle all the things that Thomas Edison invented.

Galileo

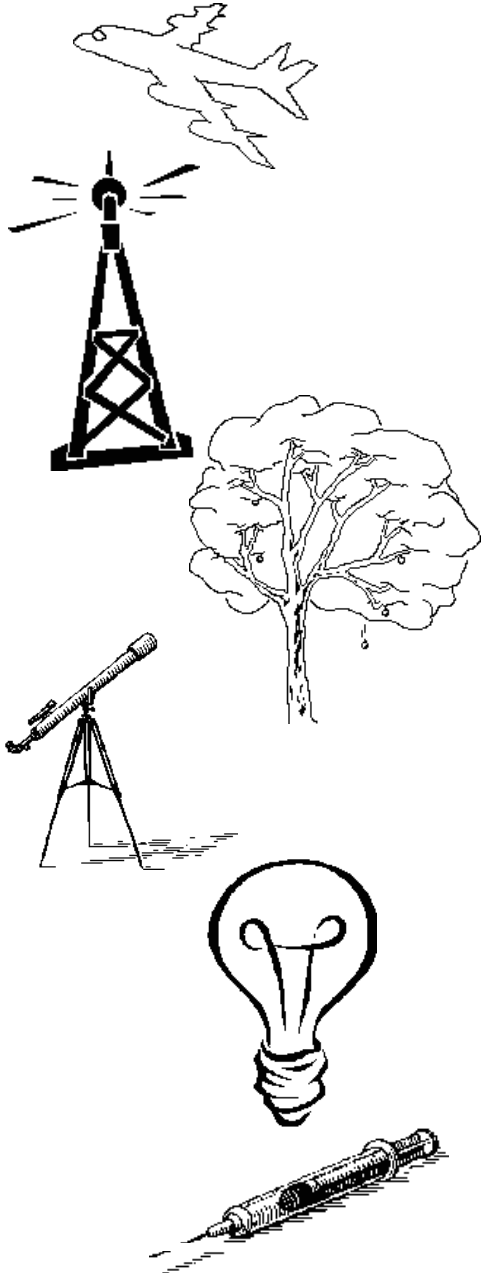
Sir Isaac Newton

Marconi

Jonas Salk

Thomas Edison

Orville and Wilbur Wright



Contributions of Scientists Overview

Draw a line from the scientist to their research.

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